
Autonomous Production Choke Controller

Submission Readiness Report

Honeywell Hackathon 2026

Engineering Submission

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Table of Contents

1. Deliverable Summary	3
2. Folder Map	3
3. Compliance	4
4. Consistency	5
5. Issues	5
6. Recommendation	5

1. Deliverable Summary

Table 1:

Category	Count	Status
Official Honeywell Deliverables	20	ALL COMPLETE
Supporting Deliverables	7	ALL COMPLETE
Python Source Files	19	ALL PRESENT
Generated Plots & Diagrams	16	ALL PRESENT
Professional PDF Reports	11	ALL PRESENT

2. Folder Map

- 01_Source_Code: 19 files
- 02_Technical_Report: Technical_Report.pdf
- 03_Presentation: presentation.md
- 04_Streamlit_Dashboard: dashboard.py + config.toml
- 05_Open_Loop_Step_Test_Analysis: 9 plots
- 06_Dynamic_Model_Identification: 2 plots
- 07_Controller_Design: Controller_Design.pdf
- 08_Validation_Report: Validation_Report.pdf
- 09_Scenario_Results: 3 plots
- 10_Architecture_Diagrams: 2 diagrams
- 11_Workflow_Diagrams: Workflow_Diagrams.pdf
- 12_Generated_Plots: 4 plots
- 13_Reference_Dataset_Comparison: Reference_Dataset_Comparison.pdf
- 14_Project_Documentation: Project_Documentation.pdf + README.md
- 15_Execution_Guide: Execution_Guide.pdf

- 16_Example_Outputs: Example_Outputs_Guide.pdf + 4 PNGs
- 17_Final_Submission_Checklist: Submission_Checklist.pdf
- 18_Submission_Ready_ZIP: Ready

3. Honeywell Requirement Compliance

Table 3:

Requirement	Status
Simulator data for identification	COMPLIANT
FOPDT from experiments only	COMPLIANT
Reference dataset not for design	COMPLIANT
Defense-in-depth safety	COMPLIANT
Constraint enforcement	COMPLIANT
Ramp rate limits	COMPLIANT
3 validation scenarios	COMPLIANT
All 6 plot types	COMPLIANT
Reproducible (seed=42)	COMPLIANT

4. Consistency Verification

- Controller implementation matches design documentation.
- Validation results match generated plots.
- Technical report references correct architecture.
- Data provenance is consistently documented across all reports.
- All documents reference the same Honeywell problem statement.

5. Remaining Issues

None. All deliverables complete, verified, and internally consistent.

6. Go / No-Go Recommendation

RECOMMENDATION: GO FOR SUBMISSION